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# **Stabilization of Expansive Soil Using Ground Granulated Blast Furnace Slag (GGBS) as a Smart Material**

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## **I. Abstract**

Expansive soils present serious challenges in civil engineering due to their high swelling and shrinkage characteristics caused by moisture variation. These soils often result in pavement distress, foundation cracking, and differential settlement. Soil stabilization is a widely adopted technique to improve the engineering behavior of such problematic soils. This paper investigates the effectiveness of Ground Granulated Blast Furnace Slag (GGBS), an industrial by-product of the steel industry, as a smart and sustainable stabilizing material for expansive soil. Laboratory investigations were carried out on untreated soil and soil stabilized with varying percentages of GGBS. The results show a significant reduction in plasticity and swelling characteristics, along with a notable improvement in strength parameters such as Unconfined Compressive Strength (UCS) and California Bearing Ratio (CBR). The study concludes that GGBS is an efficient, eco-friendly, and cost-effective smart material for stabilizing expansive soils used in pavement and foundation applications.

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## **II. Keywords**

Expansive soil, GGBS, soil stabilization, smart materials, expansive soil, sustainable construction

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## **III. INTRODUCTION**

Expansive soils, commonly referred to as black cotton soils in India, exhibit considerable volume changes due to fluctuations in moisture content. These soils contain clay minerals such as montmorillonite, which possess high water absorption capacity. The swelling during wet seasons and shrinkage during dry seasons cause severe damage to civil engineering structures, including pavements, buildings, and retaining structures.

Conventional stabilization methods using lime and cement are effective; however, they are associated with higher costs and environmental concerns due to carbon emissions. In recent years, the utilization of industrial by-products for soil stabilization has gained importance from a sustainability perspective. Ground Granulated Blast Furnace Slag (GGBS), a by-product of the iron and steel manufacturing process, exhibits latent hydraulic properties. When mixed with soil in the presence of moisture, GGBS forms cementitious compounds that enhance soil strength over time. Due to its time-dependent improvement of soil properties without continuous external intervention, GGBS can be categorized as a smart material in civil engineering applications.

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## **IV. MATERIALS USED**

The materials used in this study include expansive soil, Ground Granulated Blast Furnace Slag (GGBS), and potable water. The expansive soil was collected from a local site and air-dried, pulverized, and sieved before testing. GGBS used in this study was obtained from a steel manufacturing plant.

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## **V. METHODOLOGY**

The experimental program was carried out in the following steps:

1. Collection and preparation of expansive soil samples
2. Determination of index properties of untreated soil
3. Mixing of soil with varying percentages of GGBS (0%, 2%, 4%, 6%, 8%, 10% and 12%)
4. Compaction and preparation of test specimens
5. Curing of specimens for specified durations
6. Conducting laboratory tests on untreated and treated soil samples

Laboratory tests such as Atterberg limits, Free Swell Index, Standard Proctor test, Unconfined Compressive Strength test, and California Bearing Ratio test, Triaxial test were conducted as per relevant IS 2720 specifications. And wet-sieve analysis to classify the soil as per ISSCS.

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## VI. RESULTS AND DISCUSSION

The experimental results demonstrate that the addition of GGBS significantly improves the engineering behavior of expansive soil. A continuous decrease in liquid limit and plasticity index was observed with an increase in GGBS content, indicating reduced plasticity and swelling potential. The Free Swell Index showed a considerable reduction, confirming effective control of expansive behavior.

Strength characteristics such as UCS and CBR values increased with the addition of GGBS and with longer curing periods. This improvement is attributed to the formation of cementitious compounds such as calcium silicate hydrate (C–S–H) and calcium aluminate hydrate (C–A–H), which bind soil particles and densify the soil structure. An optimum GGBS content was observed, beyond which marginal improvement in strength was noted.

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## VII. CONCLUSIONS

Based on the experimental investigation conducted in this study, the following conclusions are drawn:

1. GGBS significantly reduces the Liquid limit and swelling behavior of expansive soil.
2. Strength parameters such as UCS and CBR increase with increasing GGBS content and curing time.
3. The stabilization mechanism is governed by pozzolanic and hydraulic reactions leading to the formation of cementitious gels.
4. GGBS acts as a smart and sustainable stabilizing material due to its time-dependent strength development.
5. The use of GGBS is economical, eco-friendly, and suitable for subgrade and foundation applications in expansive soil regions.

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